

Fall '24 — B

(Dane) (4)

①

32.99.125.255/17

72.0

$$2^{15} - 2 = 32766$$

Ans.

①

32.99.0 00000000 / 17

one
lost
address

32.99. ~~125~~ 0 / 17 N/A

32.99.125.255 / 17 → B/A Ans.

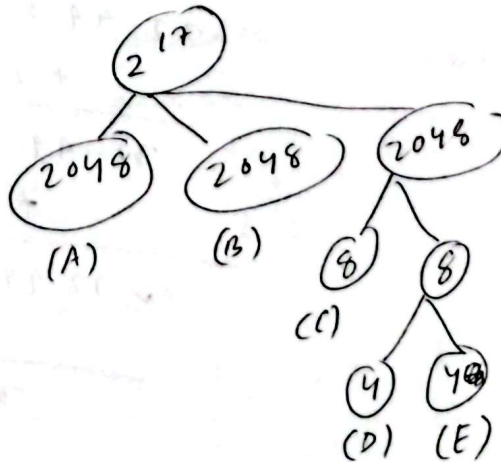
②

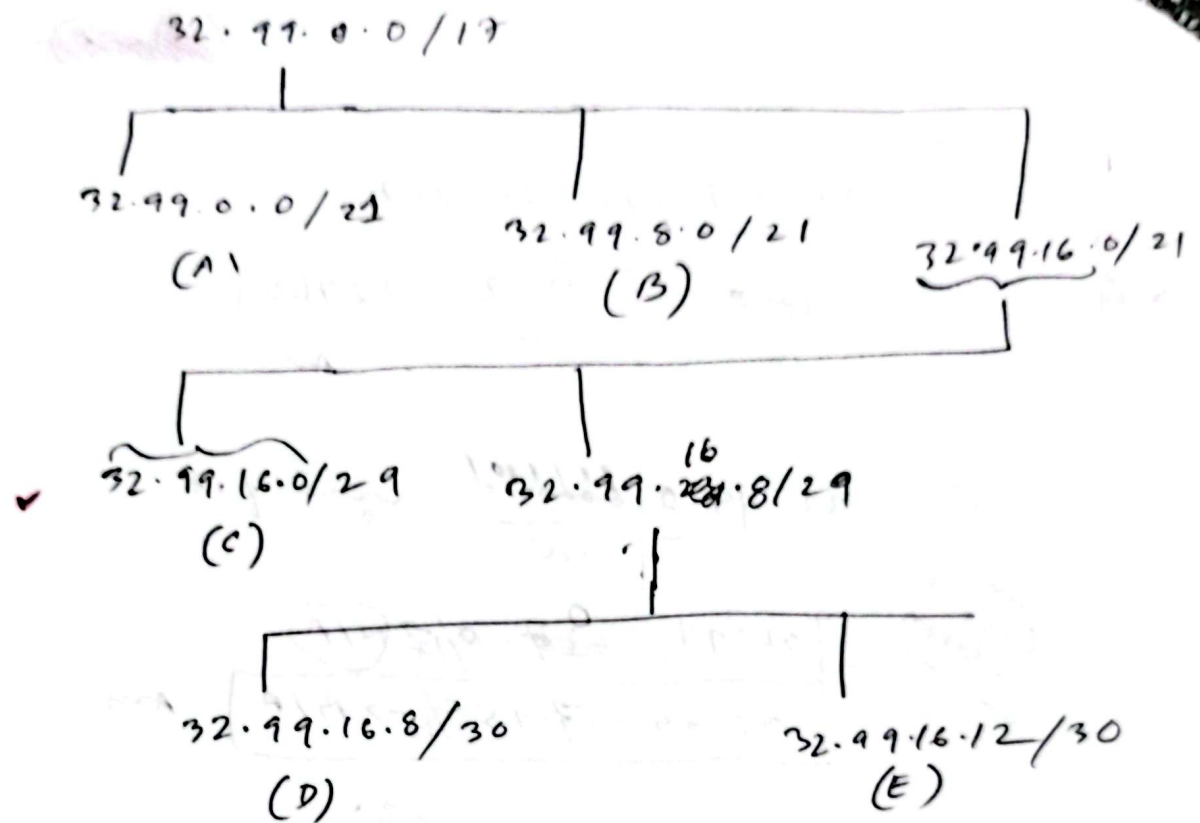
①

	H	+2	IP	Host	Net
(A)	2000	2002	2048	11	21
(B)	1024	1026	2048	11	21
(C)	3	5	8	3	29
(D)	2	4	4	2	30
(E)	2	4	4	2	30

WAN

32.99





$$32.99.0.0 \rightarrow (8 \times 256)$$

$$+ 2048$$

$$32.99.8.0$$

$$+ 2048$$

$$\checkmark 32.99.16.0$$

$$+ 8$$

$$32.99.16.8$$

$$+ 4$$

$$32.99.16.12$$

(3)

~~Header = 6~~ IHL = 6 Header = 6×4

Total length = 6421 = 24 bytes

Offset = 0 (Data + Header) (Given also)

Q209. $\frac{MTU - \text{Header}}{\text{Data size}} = 1624 \text{ bytes}$

Total Data = $6421 - 24$
 $= 6397$

no. of fragments = $\left\lceil \frac{6397}{1624} \right\rceil$
(over Data to over) = $\lceil 3.94 \rceil = 4$

$$(ii) \quad 3 \times 1624 = 4872$$

$$6397 - 4872 = 1525 \text{ bytes}$$

$$(iii) \quad \begin{array}{l} \text{min} \nearrow \\ \nwarrow \end{array} 3 \times 1624 = 4872 \quad \frac{4872}{8} = 609 .$$

(offset-63 data count always)

Fall '24 — A

(4)

(1)

✓ 1
✓ 3
→ 4

$$2^x - 2 = 16352$$

$$2^x = 16354$$

$$x = 14$$

→ host bit

$$N/Bit = 18$$

$$D/A \rightarrow 42 \cdot 1 \cdot 63 \cdot 255 / 18$$

$$(1) \quad [255 \cdot 255 \cdot 192 \cdot 0 \rightarrow S/M]$$

$$11/000000$$

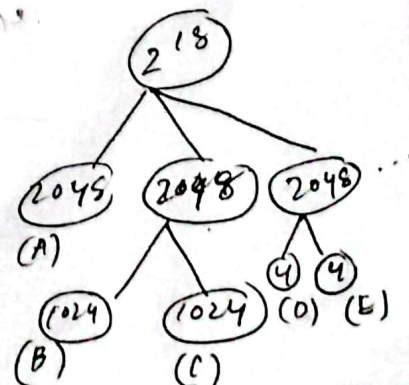
$$42 \cdot 1 \cdot 0 / 111111 \cdot 255$$

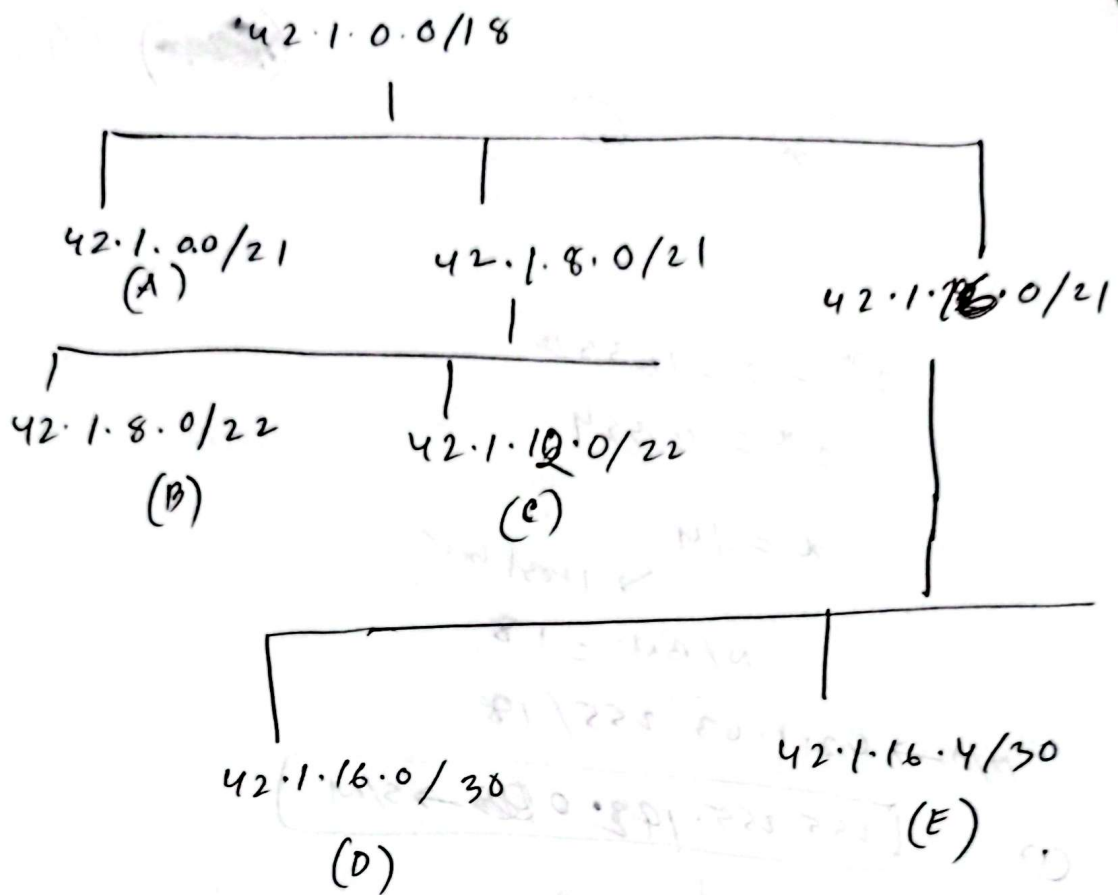
$$(1) \quad [42 \cdot 1 \cdot 0 \cdot 0 / 18] \quad N/A$$

(11)

	+2	11	Host	Net
(A) 2000	2002	2048	11	21
(B) 1022	1024	1024	10	22
(C) 512	514	1024	10	22
(D) 2	4	4	2	30
(E) 2	4	4	2	30

$$42 \cdot 1 \cdot 0 \cdot 0 / 18$$





$$\begin{array}{r}
 42.1.0.0 \\
 + 2048 \rightarrow (8 \times 256) \\
 \hline
 42.1.8.0 \\
 + 1024 \\
 \hline
 42.1.12.0 \checkmark \\
 + 1024 \\
 \hline
 \cancel{42.1.16.0}
 \end{array}$$

$$\begin{array}{r}
 42.1.16.0 \\
 + 4 \\
 \hline
 42.1.16.4
 \end{array}$$

(3)

$$IHL = 5 \quad 5 \times 4 = 20 \text{ bytes Header (given)}$$

$$\text{offset} = 0$$

$$\text{Total length} = 5086$$

Data in each fragment -

$$= 1244 \text{ ~~bytes~~ } - 20 = 1224$$

$$\text{Total data} = 5086 - 20 = 5066$$

$$(i) \quad \left\lceil \frac{5066}{1224} \right\rceil = \lceil 4.13 \rceil = 5$$

$$(ii) \quad 4 \times 1224 = 4896$$
$$5066 - 4896 = 170 + 20 = 190$$

$$(iii) \quad 4 \times 1224 = 612$$

(n-1)

Spring '24 — B

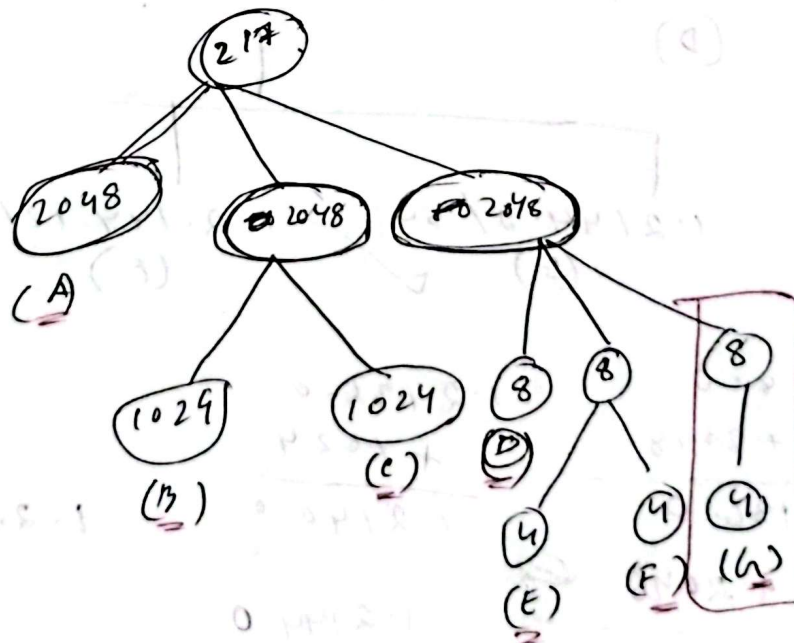
Done

✓
3

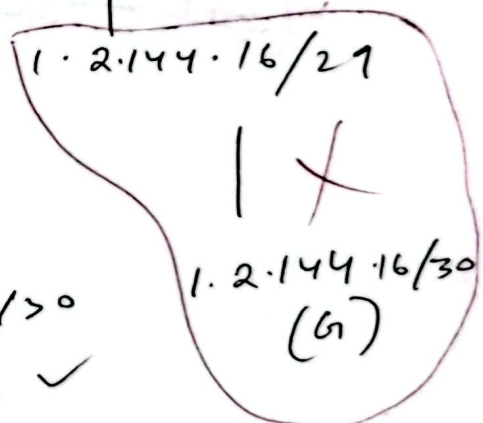
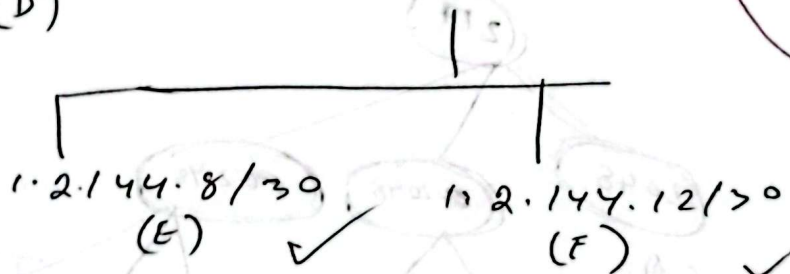
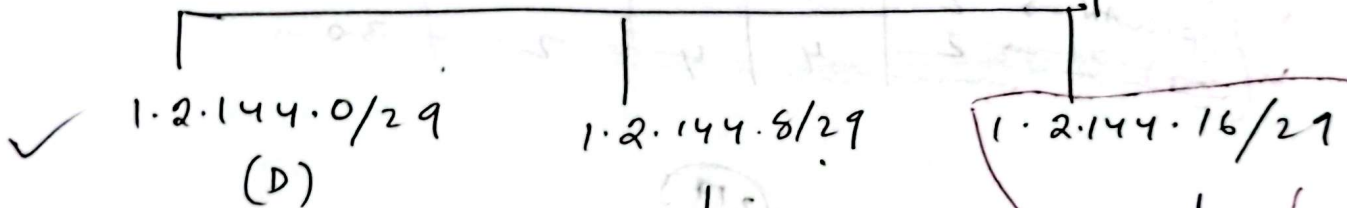
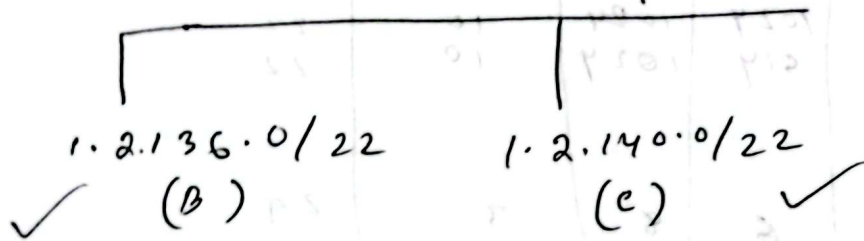
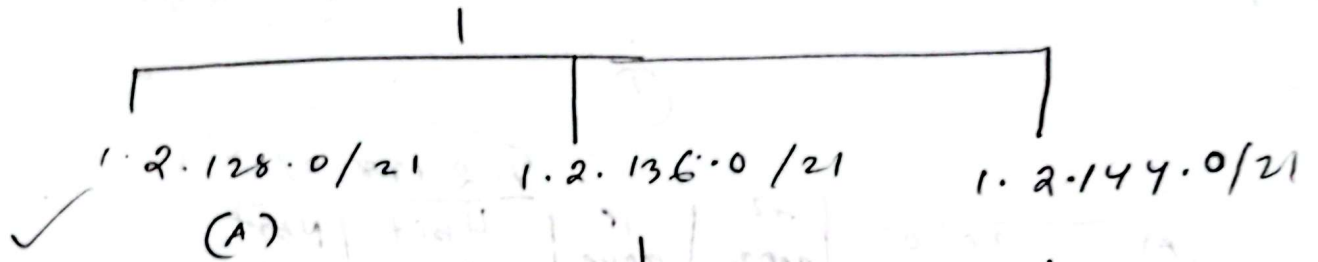
①

4. 2. 128. 0/17

		+2	18	Unit	Unit
(A)	2000	2002	2048	11	21
(B)	10 22	7024	10 24	10	22
(C)	5 12	514	10 24	10	22
	WAN → 2				
	→ 2				
(D) 52 → 2		8	8	3	29
(E) 53 → 2		4	4	2	30
(F) WAN → 2		4	4	2	30
(G) 51 → 2		4	4	2	30



1.2.128.0/17



$$\begin{array}{r} 1.2.128.0 \\ + 2048 \\ \hline 1.2.136.0 \\ + 2048 \\ \hline 1.2.144.0 \end{array}$$

$$\begin{array}{r} 1.2.136.0 \\ + 1024 \\ \hline 1.2.140.0 \end{array}$$

$$\begin{array}{r} 1.2.144.0 \\ + 8 \\ \hline 1.2.144.8 \\ + 8 \\ \hline 1.2.144.16 \end{array}$$

$$\begin{array}{r} 1.2.144.8 \\ + 8 \\ \hline 1.2.144.12 \end{array}$$

(3)

packet = 4560

header = 20

data = 4540

MTU = 380

data/MTU = 360

$$(i) \quad \left\lceil \frac{4540}{360} \right\rceil = \lceil 12.6 \rceil = 13 \quad \checkmark$$

$$(ii) \quad \frac{3 \times 360}{8} = \boxed{135} \quad \checkmark$$

offset $\rightarrow$ data always

(iii) MF = 1

①

1.2.128.0/17

Host bit $\rightarrow 15$

minimum hosts to create a network = 2

10

binary $\Rightarrow 2$ bits

can lend other 13 bits ^{to network portion} ~~subnet~~ at least a network is created.

Subnet ID

$$2^{13} = 8192$$

maximum ^{subnet} net

no. of fragments

$$\frac{(n-1) \times (\text{data/frag}) + \text{base offset}}{8}$$

8

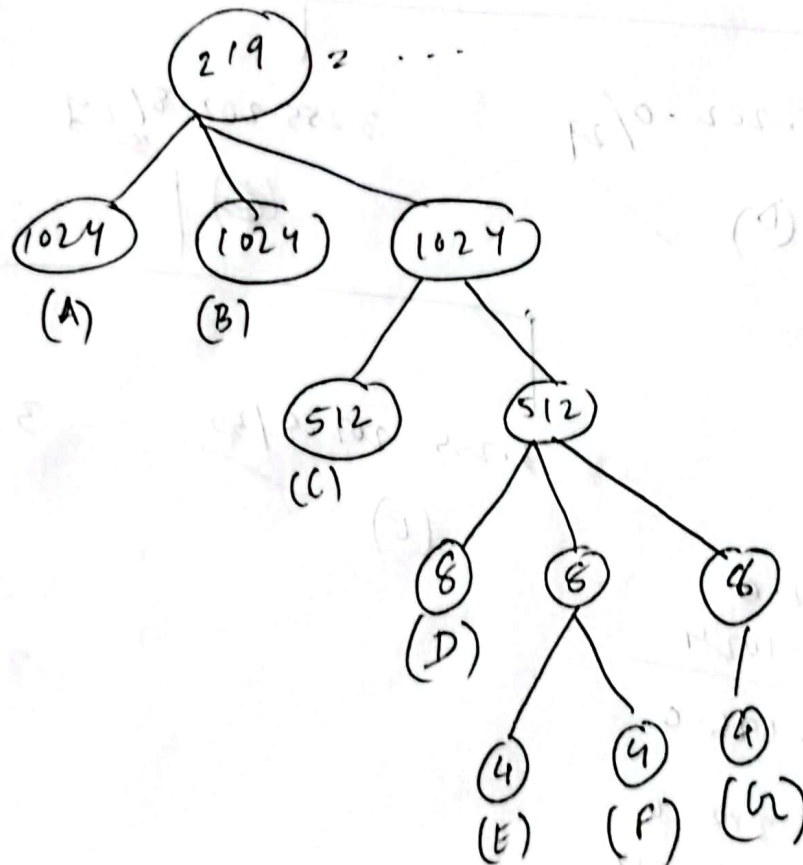
Spring '24 - A

Done
DEF

✓
✓
✓

		72	72	Hbit	Unit
(A)	1000	1002	1024	10	22
(B)	512	514	1024	10	22
(C)	255	257	512	9	23
(D)	51, 52, 3	5	8	3	21
(E)	WAN(1) 2	4	4	2	30
(F)	WAN(2) 2	4	4	2	30
(G)	53 2	4	4	2	30

3.255.192.0/19



3.255.192.0/19

3.255.192.0/22

(A) ✓

3.255.196.0/22

(B) ✓

3.255.200.0/22

3.255.200.0/23

(C) ✓

3.255.202.0/23

~~3.255.204.0/23~~

3.255.202.0/29

(D) ✓

3.255.202.8/29

~~(E)~~

3.255.202.16/29

3.255.202.8/30

(E) ✓

3.255.202.12/30

(F) ✓

3.255.192.0

+ 1024

3.255.196.0

3.255.200.0

+ 512

3.255.202.0

3.255.202.16/30

(G) ✓

(3)

195
1

data = 4080

MTU = 540

data/MTU = 520

header = 20 bytes

$$(i) \left\lceil \frac{4080}{520} \right\rceil = \left\lceil 7.86 \right\rceil = 8 \checkmark$$

$$(ii) \frac{8 \times 520}{8} = \frac{4160}{8} = 520 \checkmark$$

$$(iii) MF = 1 \checkmark$$

Spring '23 — B

4(a)

✓ 1(a)
✓ 1(b)
✓ 2(b)
→ 4(a)

$$2^n - 2 \geq 2045$$

$$2^n \geq 2047$$

$$n = 11$$

basically 2^n max capability
@ 2048 assign system

(b) $2^{11} - 2 = 2046$

(a) $109.0.8.253/21$

$$109.0.0000 \quad | \quad 000 \cdot 253$$

11 1 111111111

$$255.255.248.00$$

$$109.0.15.255/21$$

0

$$109.0.8.0/21$$

N/A

2(b)

data = 900

MF = 0

last

total 13 frag

MTU = 1945

header = 25

original datagram

Total data + header.

$$(12 \times 1920) + 900 + 25$$

$$23965 = 23965 \checkmark$$

(ii)

12th packet -

$$\frac{11 \times 1920}{8}$$

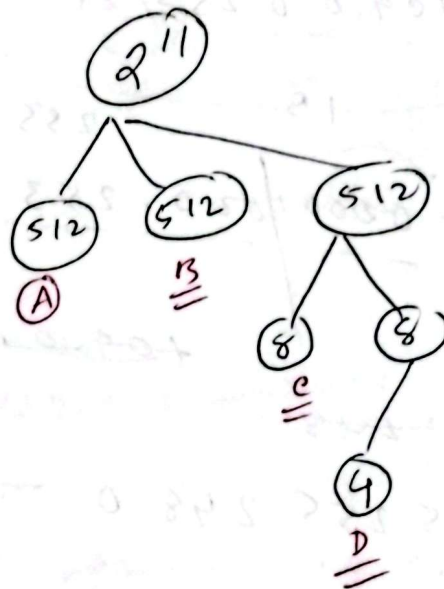
$$= 2640 \checkmark$$

(iii) MF = 0

1(b)

109.0.8.0/21

		+2	IP	Host	Net
(A)	500	502	5102	9	23
(B)	255	257	512	9	23
(C)	4	6	8	3	29
(D)	2	4	4	2	30



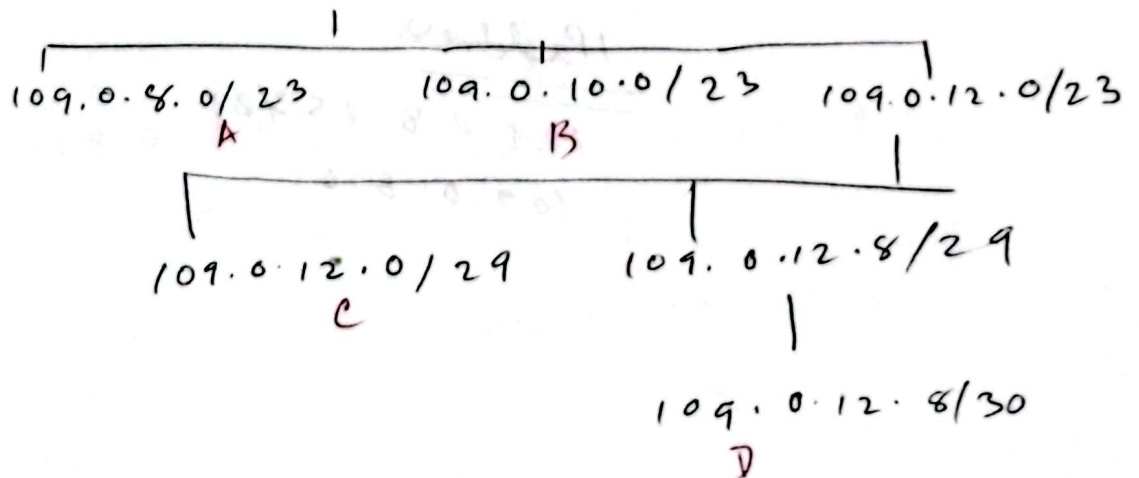
109.0.8.0
 + 512

 109.0.10.0
 + 512

 109.0.12.0
 + 8

 109.0.12.8

109.0.8.0/21



Spring '23 — A

Done

1(a) (b) ✓
2(b)
✓

109.64.0.1
109.127.255.254

01000000
1111111

(a) $[109.64.0.0/10]$ N/A

~~255.64.0.0~~
 $[255.192.0.0]$ S/M

(b) $2^{22} - 2 = 4194302$ ✓

2(b)

MTU = 955 → data = 950

Total packets/fragment = 11

MTU = 2883

head = 35

Total
payload
=

$2883 - 35 = 2848$

$(2848 \times 10) + 950$

$= [29430]$ ✓

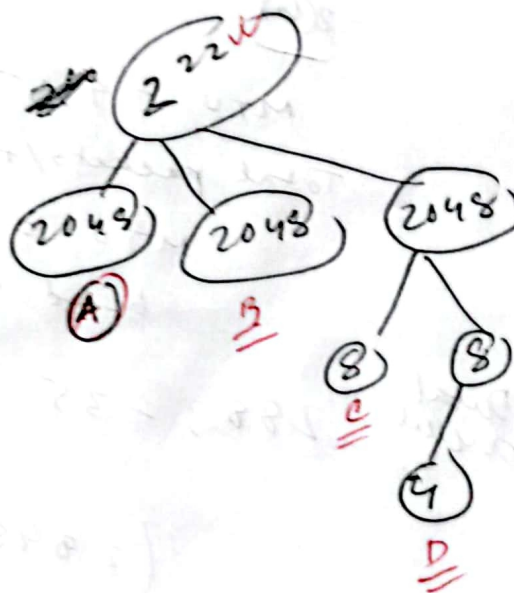
(11)
$$\frac{1 \times 2848}{8} = 356 \checkmark$$

(11) MF = 0

1(b)

	+	r	Unit	Unit
(A) 2000	2002	2048	11	21
(B) 1400	1402	2048	11	21
(C) 4	86	8	3	29
(D) 2	4	4	2	30

109.64.0.0/10



109.64.0.0/10

109.64.0.0/21

(A) ✓

109.64.8.0/21

(B) ✓

109.64.16.0/21

109.64.16.0/29

(C) ✓

109.64.16.8/29

109.64.16.8/30

(D) ✓

109.64.0.0

+ 2048

109.64.8.0

109.64.16.0

+ 8

109.64.16.8

Fall '22 — B

4(a)

- 1(a) ✓
- 1(b) ✓
- 2(b) ✓
- 4(a)

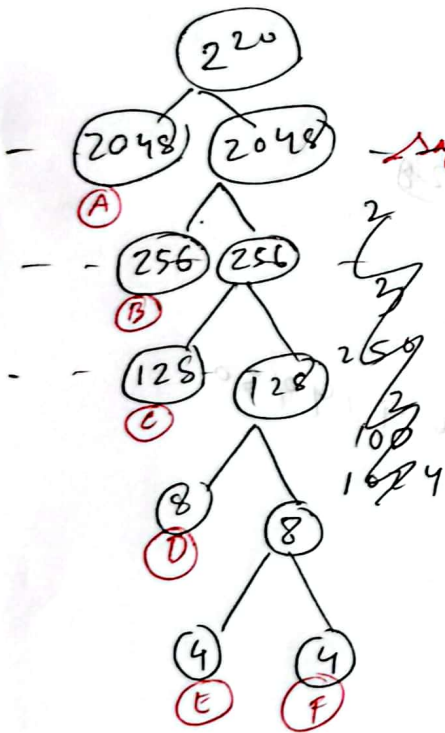
200 · 100 · 50 · 25 / 12

200 · 0 · 100 · 50 · 25 / 12

200 · 96 · 00 / 12 ✓

254

200 · 96 · 0 · 253 / 12

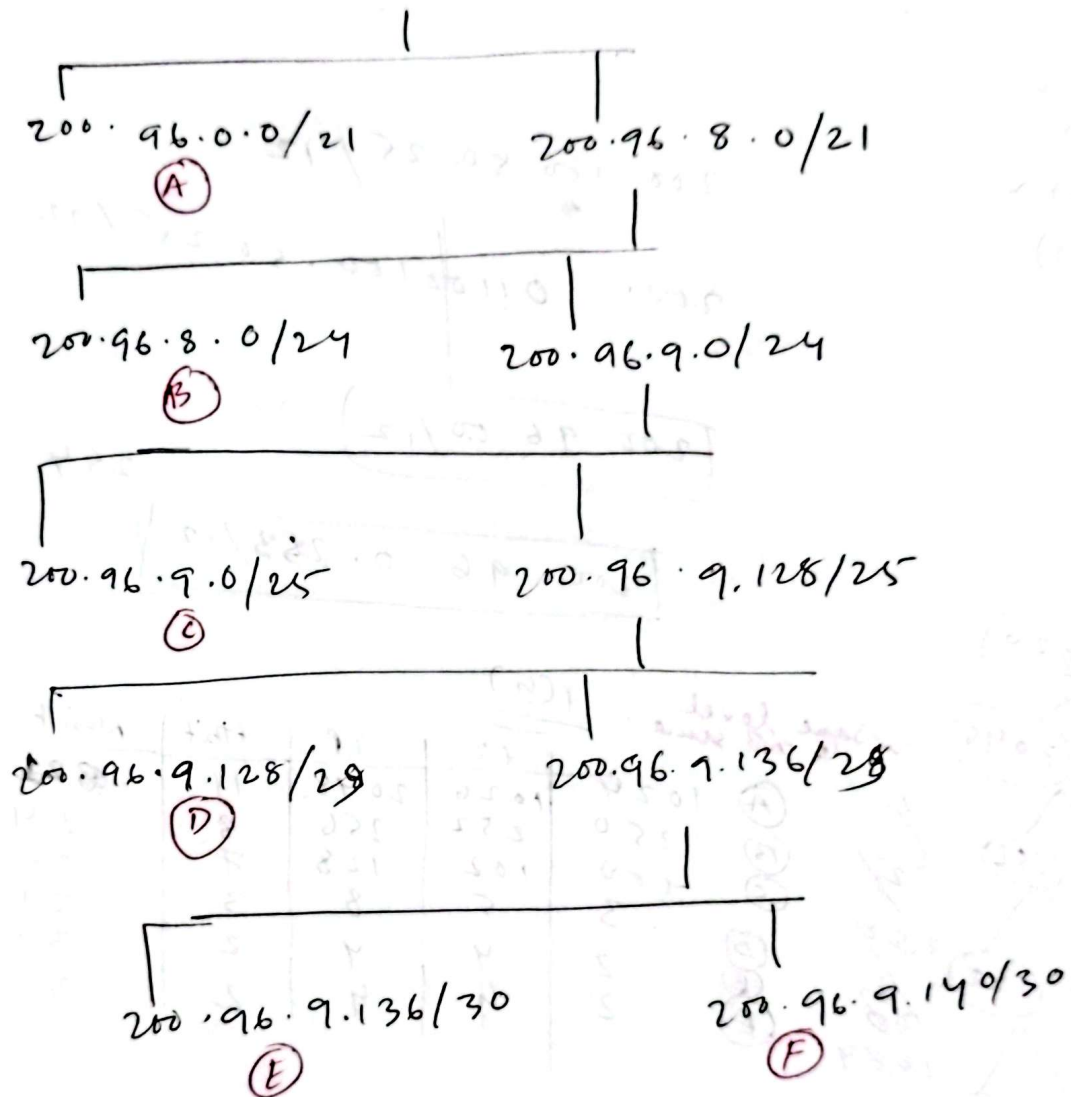


same level & not same. 1(b)

		+2	IP	Unit	Unit
(A)	1024	1026	2048	11	21
(B)	250	252	256	8	24
(C)	100	102	128	7	25
(D)	3	5	8	3	29
(E)	2	4	4	2	30
(F)	2	4	4	2	30

200 · 96 · 0 · 0 / 12

8. SS' 167



200.96.0.0

+ 2048

200.96.8.0

+ 256

200.96.9.0

+ 128

200.96.9.128.0

200.96.9.128

+ 8

136

+ 4

140

(2)

7th 268 packet-
header = 20

186

data/MTU = 248

(8) → offset-

$$(i) (248 \times 7) + 20 = \boxed{1756} \checkmark$$

(ii)

MTU = 268 bytes

(n-1)

data/MTU

Base

$$\frac{(4 \times 248) + 8}{8}$$

$$= \boxed{125} \checkmark$$

(iii) MF = 0

Fall '22 — A

4(a)

1(a) (b) ✓

2(b) ✓

→ 4(a)

(i)

$$17.50 \cdot \overset{127}{\overbrace{0 \mid 1100000}^{127}} \cdot 255 \rightarrow \textcircled{B/A} \checkmark$$

$$\overline{0 \mid 111111}$$

$$\boxed{17.50.96.81/17}$$

$$\begin{array}{r} 255.255.128.0 \\ 255.255.1000000.0 \\ \hline \textcircled{17} \end{array}$$

~~17.50.255.255/17~~

(ii)

$$17.50.0.0/17 \rightarrow \textcircled{N/A}$$

$$\boxed{17.50.0.2/17}$$

2(b)

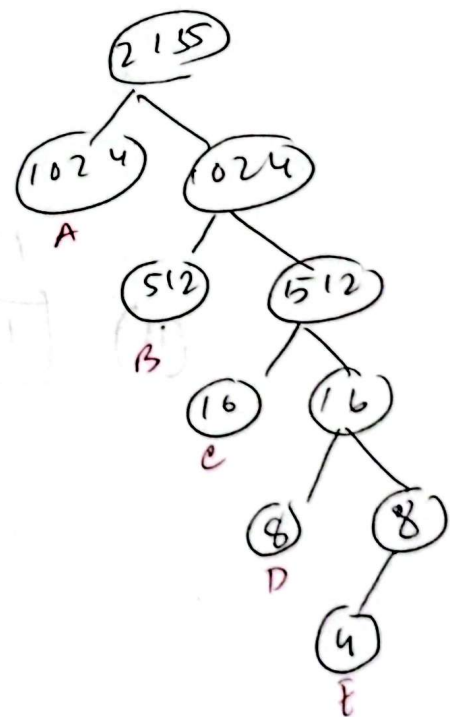
			+2	+1	Host	Net
2	A	512	514	1024	10	22
4	B	255	256	512	9	23
8	C	13	15	16	4	28
16	D	4	6	8	3	29
32	E	2	4	4	2	30

254+1

=255

(default gateway)

17.50.0.0/17



$$17.50.0.0/17$$

$$17.50.0.0/22 \quad 17.50.4.0/22$$

(A)

$$17.50.4.0/23 \quad 17.50.6.0/23$$

(B)

$$17.50.6.0/28 \quad 17.50.6.16/28$$

(C)

$$17.50.6.16/29 \quad 17.50.6.24/29$$

(D)

$$17.50.6.24/30$$

(E)

$$17.50.0.0$$

$$+ 1624$$

$$17.50.4.0$$

$$+ 512$$

$$17.50.6.0$$

$$+ 16$$

$$17.50.6.16$$

2(b)

(8th
unit)

packet = 230

182 off

header = 22

$$\rightarrow \text{data/MTU} \\ (8 \times 208) + 22$$

(i)

= 1686 ✓

(ii)

$$\boxed{230} = \text{MTU} \\ \checkmark$$

$$\frac{(5 \times 208) + 0}{8} = \boxed{130} \checkmark$$

(iii)

MF = 1

Summer '22 — B

4(c)

$(a)(b)$ ✓
 $\rightarrow 4(c)$
 $5(b)$ ✓

(1)

19.253.99.49/15

255.254.0.0

255.11111110.00

19.11111101.99.45

(1)

19.253.255.255 B/A ✓

19.11111101.0.0.0 $\rightarrow$ (N/A)

19.252.0.0/15

19.255.0.2/15

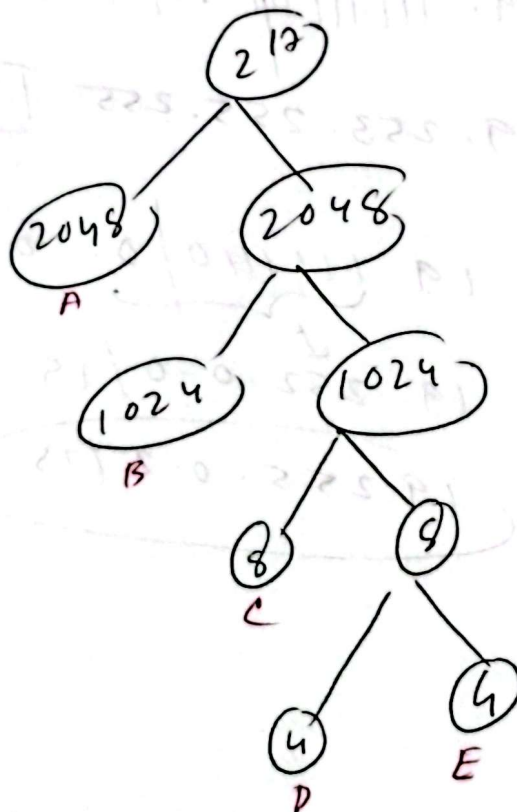
SV

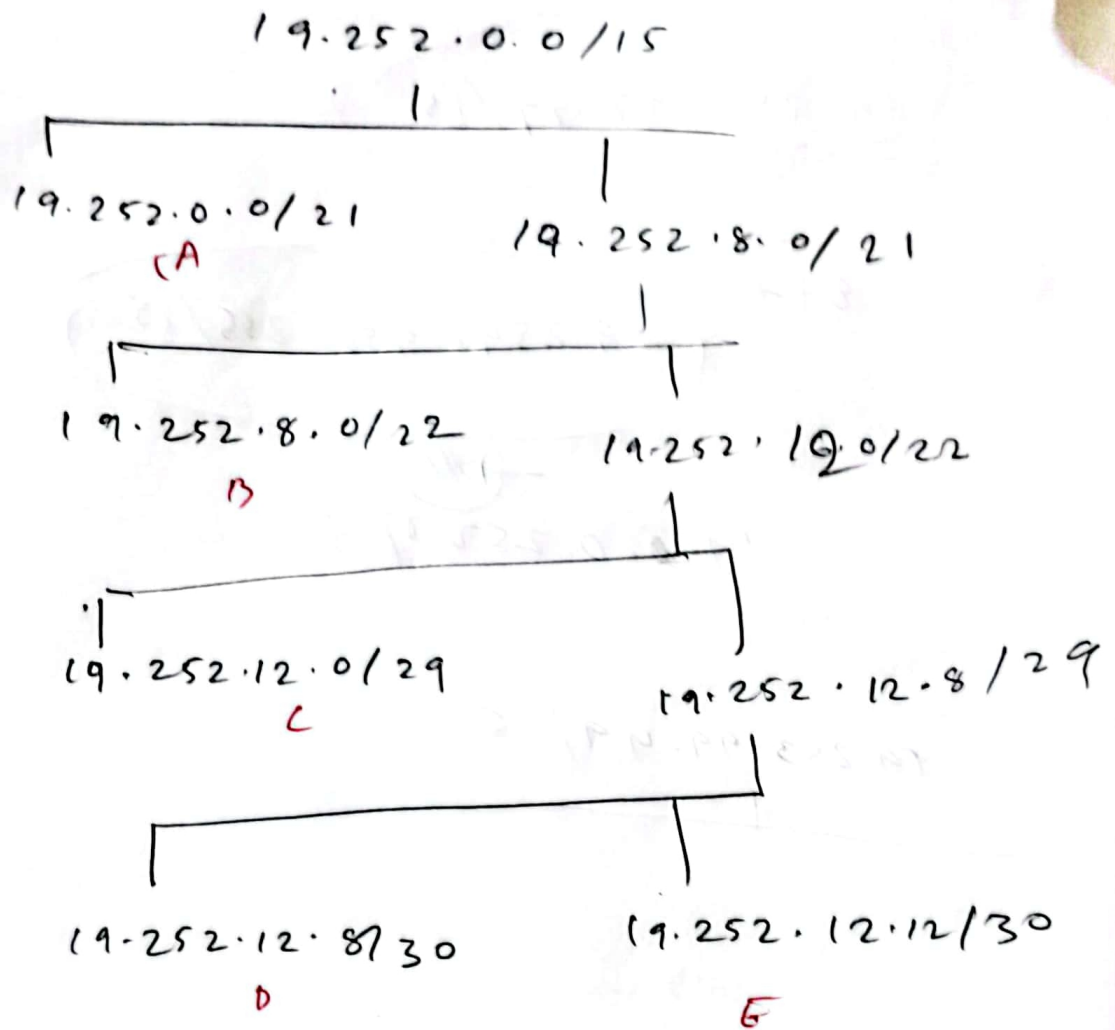
RF 10 : RF : : AC 19 : 0 : 0 : E000

(b)

		+2	IP	Height	Count
511	(A) 1240	1242	2048	11	21
1240	(B) 511	513	1024	10	22
2	(C) 3	5	8	3	29
2	(D) 2	4	4	2	30
3	(E) 2	4	4	2	30

19.252.0.0/15





$$\begin{array}{r}
 19.252.0.0 \\
 + 2048 \\
 \hline
 19.252.8.0 \\
 + 1024 \\
 \hline
 19.252.12.0
 \end{array}$$

Summer '22 — A

4(a)

1(a) ✓

1(b) ✓

→ 4(a)

✓ 5(b) FF10 : AC19 : 0 : 1000 : 6000

19 · 96 · 99 · 49 / 10

to

19 · 01 / 100000 · 99 · 49

 000000

(i) 19 · 64 · 0 · 0 / 10 → (NA)

(ii) 255 · 192 · 0 · 0

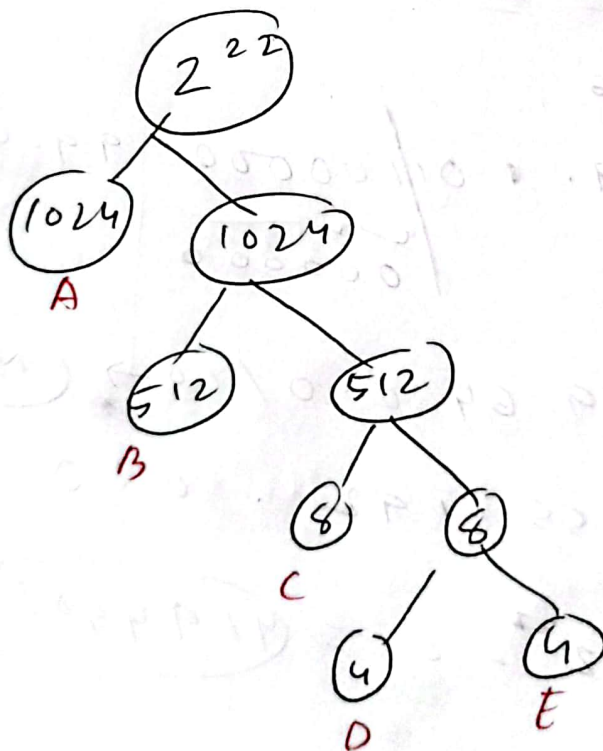
(iii) $2^{22} - 2 = 4194302$ ✓

255

1(6)

		+2	IP	Host	Sub
2	A 600	602	1024	10	2
2	B 255	257	512	9	23
3	C 3	5	8	3	29
	D 2	4	4	2	30
600	E 2	4	4	2	30

19.64.0.0/10



$$\begin{array}{r} 19.64.0.0 \\ + 1024 \\ \hline 19.64.4.0 \end{array}$$

19.64.0.0/10

